

CHAPTER 1:

INTRODUCTION

Effective communication and real-time soldier monitoring are critical requirements in combat zones, disaster rescue missions, and remote security operations. In many field conditions, traditional communication systems such as mobile networks, radio towers, or satellite links are unreliable due to distance, terrain obstacles, or intentional jamming. TACMESH-VITAL: Smart and Secure Mesh Network for Battlefield Soldier Monitoring and Ammo Tracking addresses these challenges by creating a fully decentralized, wearable monitoring system using multiple ESP32 nodes.

Each wearable node continuously records biometric data using the MAX30105 sensor, including heart rate, oxygen saturation (SpO₂), and body temperature. It also tracks GPS location, counts shots fired using a trigger button, and displays relevant information on an OLED screen with silent vibration alerts. Instead of relying on a central router, the nodes communicate through ESP-NOW mesh networking, which enables direct device-to-device data exchange and automatically forwards messages across multiple hops. This forms a self-healing tactical mesh network that remains functional even if some nodes move, disconnect, or fail.

The system also estimates inter-node distances using RSSI (Received Signal Strength Indicator), allowing soldiers to determine approximate proximity without needing constant GPS visibility. Nodes with Wi-Fi connectivity act as gateways, sending telemetry to a Node.js WebSocket dashboard. The dashboard provides a real-time view of soldier vitals, ammo status, connectivity, network structure, signal strength, and emergency alerts, making it easier for commanders to monitor the field and issue instructions directly to specific soldiers or all nodes.

A core feature of TACMESH-VITAL is the emergency response mode, where any soldier can instantly broadcast a distress signal. Nearby nodes then receive direction and distance guidance, enabling fast and coordinated responses. By combining embedded sensing, wireless mesh communication, and interactive visualization, TACMESH-VITAL improves battlefield awareness, enhances decision-making, and supports faster medical assistance and tactical movement in critical environments.

CHAPTER 2:

LITERATURE SURVEY

Real-time soldier monitoring and tactical communication systems have been a major research focus in military electronics and IoT-based safety systems. Many researchers have attempted to integrate wireless communication with physiological sensing to track soldier health and location during missions. Existing literature typically combines GPS for geographical positioning, biosensors for health parameters, and wireless modules for communication to a remote base station. However, most previous systems rely on a centralized architecture, limited connectivity options such as GSM or single-hop communication, and do not address the need for multi-node interaction, emergency assistance, or tactical data like ammunition tracking. To understand the current state of technological solutions, several research works related to GPS-based location tracking, health monitoring using wearable sensors, and wireless communication technologies have been reviewed. These studies highlight both the progress achieved and the limitations that still exist in battlefield monitoring systems.

Maria Manoj B. et al. (2023) in their paper titled “IoT Based Military Health Service in Battle Field and GPS Tracking” propose a soldier monitoring unit that measures heart rate and body temperature using biosensors and tracks GPS location through an IoT module. The authors use a PIC16F877A microcontroller to collect sensor data and transmit it to a server unit along with latitude and longitude information. The system provides alerts when vital values cross threshold limits and activates a Peltier cooling or heating mechanism to protect soldiers in extreme weather conditions. Communication is mainly based on IoT and the data is displayed on an LCD and monitored at a base station, with an emergency button for distress messaging. Compared to this system, our project TACMESH-VITAL adds advanced tactical features such as ESP-NOW mesh networking, RSSI-based distance estimation, ammo tracking, vibration alerts, and real-time dashboard visualization. Instead of relying on a single controller and simple IoT transmission, TACMESH-VITAL supports multi-hop soldier-to-soldier communication, command messages, OLED display feedback, GPS + mesh distance mapping, and emergency assistance navigation, making it more suitable for real battlefield teamwork and decision support [1].

Most existing soldier monitoring systems focus on basic real-time health tracking and location transmission using wireless communication. In the work by Laxman Thakre et al. (2022), a GPS and

GSM based monitoring system is presented, where soldier vital parameters such as heart rate and temperature are measured using biosensors and transmitted to a base station through SMS communication. Although this approach provides location tracking and health data, it depends entirely on GSM network availability and supports only one-way communication, where data flows from soldier to control room without command feedback. Several earlier studies also use similar architectures involving Arduino controllers, GPS modules, GSM connectivity, and LCD displays to send positional information and health status to a registered mobile number. These designs are simple, low-power and low-cost, but they suffer from major limitations such as slow data transfer, lack of mesh connectivity, absence of interoperability between multiple soldiers, no visualization dashboard, and no tactical information like ammunition status or emergency guidance. Most reported systems focus only on tracking and monitoring functions, without providing coordinated communication between soldiers or base station control. In contrast, our proposed system TACMESH-VITAL advances beyond existing solutions by introducing ESP-NOW based wireless mesh networking, real-time bidirectional command communication, RSSI-based distance estimation, dashboard visualization, ammunition tracking and emergency assistance navigation, thereby addressing the gaps identified in reviewed literature and offering more comprehensive battlefield support [2].

Anubhav Kumar Pandey et al. (2025) in their paper titled “Cyber Physical System Based Wearable Health Monitoring Device for Defense Applications” describe a wearable glove system designed for soldiers that can measure heart rate, SpO₂, body temperature, motion, and surrounding environmental conditions. The authors use a combination of ZigBee mesh networking for communication between nearby soldier devices and NB-IoT to send the collected data from a gateway to a remote control room. Their main aim is to provide continuous health monitoring in extreme military environments with low power consumption and reliable wireless transmission. The system focuses mainly on physiological sensing and health data transfer, but it still depends on a central gateway node for long-range communication. In comparison, our project TACMESH-VITAL expands this idea from only health monitoring to complete tactical and situational awareness. We use ESP-NOW multi-hop mesh networking, which allows direct soldier-to-soldier communication without a fixed gateway. In addition to measuring heart rate, SpO₂, and temperature, our system includes ammo tracking, vibration alerts, GPS location, RSSI-based distance estimation, real-time OLED display messages, and dashboard command control. This provides not just medical data, but also team coordination, emergency navigation, and faster decision-making on the battlefield, which is not covered in the work by Pandey et al [3].

Kiran Ingale et al. (2023) in their paper titled “Battlefield Health Surveillance with Soldier Tracking: A Privacy-Preserving Encrypted Approach” describe a wearable jacket system for soldiers that measures vital signs such as heart rate, body temperature, ECG, oxygen level, and GPS location. The system uses an ESP32-CAM, multiple sensors, a GSM module for SMS alerts, and cloud platforms like ThingSpeak to visualize data. A key focus of this work is data security where soldier information is encrypted to prevent unauthorized access. The system mainly concentrates on health monitoring, location tracking, and privacy protection during battlefield operations. In comparison, our project TACMESH-VITAL improves on this approach by adding ESP-NOW mesh networking, RSSI-based distance estimation, live ammunition tracking, OLED feedback, vibration alerts, and a real-time command dashboard. Unlike Ingale et al.’s system which relies on a single server and GSM, TACMESH-VITAL supports multi-hop communication without a central gateway, tactical messaging between soldiers, and emergency assistance navigation, making it suitable for real battlefield coordination and decision-making [4].

Miroslav Hopjan et al. (2021) in their paper titled “Mesh Network Application” present a survey of ad-hoc mesh networks, their basic classifications, routing methods, and military communication use cases. The paper explains how mesh networks allow every node to act as both a sender and a router, creating self-organizing, multi-hop, peer-to-peer communication without a central infrastructure. It discusses MANETs (Mobile Ad-hoc Networks), wireless sensor networks, 802.11 mesh routing protocols, delay-tolerant networking, and battlefield applications where routing adjusts automatically if a node fails or moves. In comparison, our project TACMESH-VITAL applies these mesh networking concepts practically in a soldier health and tactical monitoring system. Instead of only studying network theory, we implement ESP-NOW multi-hop communication, real-time telemetry sharing, RSSI-based distance calculation, and GPS tracking between soldiers. In addition to mesh communication, TACMESH-VITAL also provides health monitoring, ammo tracking, vibration alerts, OLED feedback, and a command dashboard, making it not only a networking study but a complete real-time field application based on mesh network principles [5].

Atif Ali et al. (2020) in their paper titled “Military Operations: Wireless Sensor Networks Based Applications to Reinforce Future Battlefield Command System” provide an overview of how Wireless Sensor Networks (WSNs) can be used in modern battlefield communication and soldier tracking. The authors describe different types of sensors used in military environments such as acoustic sensors, presence sensors, chemical detection sensors, and sniper localization sensors. They explain that **WSNs** allow self-organizing, multi-hop communication, where sensor nodes can collect and forward data

across a battlefield without relying on a central infrastructure. The paper focuses mainly on **sensor** classification, battlefield data collection, communication architecture, and future research challenges for military WSN systems. In comparison, our project TACMESH-VITAL takes this WSN concept and applies it directly to real-time soldier monitoring and tactical communication. While the paper discusses WSN theory and military use cases, TACMESH-VITAL implements a practical mesh network using ESP-NOW, sensor data sharing, RSSI distance estimation, GPS tracking, ammo counting, vibration alerts, and dashboard command control. This makes TACMESH-VITAL not just a theoretical study of sensor networks, but a complete and functional system for soldier assistance, emergency navigation, and field coordination [6].

Fooz Althqfiy et al. (2024) in their paper titled “Revolutionizing Remote Mission Support: A Smart Suit for Tracking and Vital Sign Monitoring” propose a wearable smart suit that measures vital signs such as heart rate and body temperature and tracks the user’s GPS location. The system uses LoRa communication to send data over long distances to a mission headquarters, where a central control room monitors soldiers’ status and receives emergency alerts. The paper shows simulation results using Proteus and Cisco Packet Tracer, and includes a dashboard where soldiers can be marked as “Fine,” “Alarm,” or “Lost” based on their vital readings and location information. In comparison, our project TACMESH-VITAL adds additional tactical features beyond health and location monitoring. We implement ESP-NOW mesh networking without a single central gateway, real-time ammo tracking, RSSI-based distance estimation, vibration alerts, OLED display messages, and a full interactive dashboard with command control. This makes TACMESH-VITAL suitable not only for remote health monitoring, but also for team coordination, emergency navigation, and battlefield communication, which is not covered in Althqfiy et al.’s work [7].

Swathi Thotakura et al. (2025) in their paper titled “LifeLine: Smart Soldier Health Monitoring and Situational Awareness System Powered by ESP8266 and IoT Integration” present an IoT-based soldier monitoring system that measures vital parameters such as temperature, blood pressure, oxygen level (SpO₂), and heart rate. The system uses ESP8266, wearable sensors, and a GPS module to send real-time data to a cloud server through Wi-Fi, enabling medical teams to observe soldier health and location remotely. The paper shows data flow diagrams and hardware results, including sensor readings from different trials (Table 1 on page 5) which demonstrate accurate detection of abnormal conditions in real time. The authors mainly focus on medical monitoring, alert generation, and remote storage of soldier health data on IoT platforms. In comparison, our project TACMESH-VITAL extends beyond basic IoT monitoring. Instead of depending only on Wi-Fi to a cloud server, we use ESP-NOW

mesh networking, real-time ammo tracking, RSSI-based distance estimation, emergency vibration alerts, OLED feedback, and a live tactical dashboard with command control. Unlike the LifeLine system which requires central IoT connectivity, TACMESH-VITAL supports multi-hop soldier-to-soldier communication, field commands, emergency acceptance, team navigation, and mesh-based distance routing, making it suitable for both medical safety and tactical battlefield coordination [8].

Madhu Sudarmani et al. (2022) in their paper titled “IoT Based Healthcare Monitoring and Tracking System for Soldiers using ESP32” propose a soldier unit combining ESP32, GPS, heart rate sensor, temperature sensor, bomb detector, and panic button. The collected health and location data is sent wirelessly to a base station and displayed through a cloud platform like ThingSpeak, enabling continuous monitoring from remote locations. The system focuses on WBAN (Wireless Body Area Network) communication and provides emergency alerts to the control unit when abnormal conditions are detected. The authors mainly address health safety, remote tracking, and emergency messaging using IoT modules and cloud services. In comparison, our project TACMESH-VITAL adds more tactical functionality beyond basic IoT monitoring. Instead of depending on a single cloud path, we implement ESP-NOW mesh networking, allowing soldier-to-soldier data transfer without internet. We also include ammo tracking, vibration alerts, OLED display feedback, RSSI-based distance estimation, emergency acceptance mode, and a real-time WebSocket dashboard for command control. This makes TACMESH-VITAL suitable for live battlefield coordination and tactical decision support, which is not covered in Sudarmani et al.’s work [9].

Jomole Varghese Vadakkan et al. (2024) in their paper titled “Quality-Aware Rhythm-Aware Heart Rate Variability Analysis for Soldier Health and Wellness Monitoring Using Wearable ECG Sensing Devices” propose an advanced ECG-based soldier monitoring system that reduces false alarms during noisy battlefield conditions. The authors use CNN models to detect irregular heart rhythms, muscle artifact detection methods, and HRV (Heart Rate Variability) analysis to improve accuracy in stress and emotion monitoring. Their system focuses mainly on ECG signal quality, rhythm classification, and noise reduction to get reliable heart rate parameters under motion or electrode disturbance. In comparison, our project TACMESH-VITAL extends beyond ECG analysis by including ESP-NOW mesh networking, GPS tracking, RSSI distance estimation, ammo tracking, OLED display alerts, vibration feedback, and a real-time dashboard. While Vadakkan et al. focus on signal processing and detection accuracy for soldier health, TACMESH-VITAL provides complete tactical communication and field coordination, making it suitable not just for monitoring vital signs but also for decision-making, emergency assistance, and real-time soldier interaction in battlefield environments [10].

Ayyanar et al. (2023) in their paper titled “Remote Tracking of UAV Swarms via 3D Mobility Models and LoRaWAN Communications” present a long-range wireless tracking system designed for multiple UAVs (drones) operating as a group. The authors use LoRaWAN, which supports low-power and long-distance communication, to continuously transfer position and telemetry data. The system applies 3D mobility models to understand drone movement in real time and track their exact location in a swarm. Their work mainly focuses on coverage, mobility planning, and reliable long-range communication for surveillance and mission support. The study shows that LoRaWAN can efficiently support **multi-node** tracking over wide geographic areas, especially when internet connectivity is not available. In comparison, our project TACMESH-VITAL applies a similar multi-node tracking idea, but instead of drones, we monitor live soldiers in a battlefield. We improve on Ayyanar et al.’s concept by using ESP-NOW mesh communication, which allows soldier nodes to communicate with each other directly without a base station. Along with location data, TACMESH-VITAL adds real-time vital sign monitoring (heart rate, SpO₂, temperature), ammunition tracking, OLED display, vibration alerts, emergency button interaction, and dashboard command control. While the UAV work focuses only on tracking movement using LoRaWAN, our system provides both health safety and tactical decision support by combining sensors, mesh networking, and soldier interaction features. This makes TACMESH-VITAL more suitable for ground combat operations, especially in areas where internet and normal communication are unavailable [11].

Dr. B. Veerasamy et al. (2023) in their paper titled “Soldier Health Detection and Position Tracking System using LoRaWAN Sensor for Low Power and Long-Range Access” present a wearable IoT device that continuously measures a soldier’s heart rate, blood oxygen level (SpO₂), body temperature, and GPS location. The system uses a LoRaWAN sensor module linked to a STM32 microcontroller to send health data to a control room over long distances with low power consumption. The authors include hardware results showing real sensor outputs and explain an emergency button feature that transmits the soldier’s location if help is needed. In comparison, our project TACMESH-VITAL upgrades this concept by replacing single-gateway LoRaWAN communication with ESP-NOW multi-hop mesh networking between soldiers. We add more tactical capabilities such as ammo tracking, RSSI-based distance estimation, vibration alerts, OLED display feedback, command messaging, and real-time dashboard visualization. Unlike the LoRaWAN system that only sends health data to a control center, TACMESH-VITAL enables soldier-to-soldier communication, emergency acceptance, navigation to injured soldiers, and distributed situational awareness, making it more suitable for **active** battlefield teamwork instead of only health monitoring[12].

Chowdam Anila et al. (2024) in their paper titled “Sun-Powered Tactical Uniforms for Soldiers in Military Applications” present a smart tactical uniform powered by solar energy to continuously monitor soldier health and environmental conditions. The authors use an Arduino microcontroller connected to sensors for heart rate, body temperature, humidity, and GPS location, and they integrate a 70×70 mm solar panel to charge a 12 V battery for powering the system. The design includes a Peltier heating and cooling module, CPU fan, GSM communication module, and LCD display to ensure thermal comfort and emergency data transmission in remote military bases. The main objective of this work is to reduce power dependency by using renewable solar energy and provide basic health monitoring and alert features for soldiers operating in extreme climate conditions. In comparison, our project TACMESH-VITAL builds on this idea by addressing not just power and health, but full battlefield communication and tactical awareness. Instead of using GSM with a single base station, we implement ESP-NOW mesh networking to allow direct soldier-to-soldier communication without cellular networks. Additionally, TACMESH-VITAL tracks ammunition usage, RSSI-based distance between soldiers, GPS position, emergency navigation, vibration alert feedback, and OLED display messages, along with a live WebSocket dashboard for command and control. While the work by Anila et al. focuses mainly on energy sustainability and simple monitoring, TACMESH-VITAL provides a complete real-time tactical system for health, movement, communication, and battlefield decision support, which makes it more suitable for active combat missions where connectivity, coordination, and fast response are critical [13].

Moni Sree H et al. (2022) in their paper titled “Smart Military E-Outfit of the Future” describe a wearable uniform system that embeds multiple sensors into military clothing to monitor a soldier’s health and environmental conditions. The outfit uses a PIC microcontroller, temperature sensor, heartbeat sensor, sweat sensor, GPS, and thermoelectric Peltier heating elements to keep soldiers warm in extreme cold, while also tracking their vital signs and location. Data is sent to Adafruit cloud using Wi-Fi, and the design focuses on comfort, flexible electronics, and heating technology in fabric. The authors mainly aim to reduce mortality due to cold weather and provide basic real-time health monitoring through smart clothing. In comparison, our project TACMESH-VITAL expands beyond wearable thermal comfort by implementing ESP-NOW mesh networking, real-time ammo tracking, RSSI-based soldier distance detection, vibration alerts, OLED tactical display, WebSocket dashboard control, and two-way communication between soldiers. Unlike the E-Outfit system which depends on cloud connectivity and focuses primarily on health and warming features, TACMESH-VITAL

provides complete tactical coordination and emergency support suitable for live battlefield operations, making it more useful for decision-making and team navigation in combat environments [14].

Patrik Kutilek et al. (2017) in their paper titled “Wearable Systems for Monitoring the Health Condition of Soldiers: Review and Application” provide a detailed review of different wearable sensors used to monitor soldier health and stress levels. The paper explains common biosensors such as heart rate, ECG, respiration, pulse oximetry, body temperature, motion tracking, and perspiration sensors, and describes how they are applied in systems like the FlexiGuard wearable monitoring unit used in the Army of the Czech Republic. The authors summarize sensor advantages, limitations, and suitability for battlefield conditions, with a focus on medical monitoring, stress detection, and sensor placement for long-term use in rough environments. In comparison, our project TACMESH-VITAL includes basic vital sensors such as heart rate, SpO₂, and temperature, but extends beyond sensor technology by adding ESP-NOW mesh networking, GPS tracking, RSSI distance estimation, ammo tracking, OLED interface, vibration alerts, and live dashboard monitoring. While Kutilek et al. focus mainly on reviewing sensors and their health applications, TACMESH-VITAL offers a complete real-time tactical communication system for soldiers, combining health monitoring with battlefield coordination and emergency response [15].

CHAPTER 3:

PROBLEM ANALYSIS AND SOLUTION

3.1 Problem Definition

In modern battlefield environments, soldiers operate in unpredictable and extremely dangerous conditions where their health and survival depend heavily on real-time information and communication. Yet, the biggest challenge is the lack of a reliable, continuous, and secure method to monitor soldiers' physical condition, location, and operational status once they move away from base camps or communication towers. Traditional systems mainly use GSM, Wi-Fi, or satellite networks, which are often unavailable, unstable, or easily disrupted due to geography, signal blocking, jamming, or equipment failure. When a soldier enters areas such as forests, deserts, tunnels, mountains, or enemy zones, network connectivity becomes weak or completely unavailable. As a result, commanders lose the ability to see who is safe, who is injured, who needs help, and where they are.

A major problem in existing military monitoring is that current systems rely **on** single-point communication architecture. Data from soldiers is normally sent to one fixed control station or server. If this station is destroyed, hacked, out of battery, or moved out of range, all communication stops. Soldiers have no way to communicate directly with each other, especially when separated during missions. This creates dangerous gaps in situational awareness. If a soldier collapses, gets shot, becomes unconscious, or is under threat, the situation remains unknown to nearby teammates because there is no automatic emergency notification method available between soldiers.

Monitoring of soldier health is another critical problem area. Existing devices may collect vital signs such as heart rate, temperature, or oxygen level, but many of them are not designed for continuous real-time battlefield usage. Some devices only record data and do not send automatic alerts. If a soldier experiences medical issues like low oxygen levels, heat stroke, severe bleeding, heart irregularities, or fatigue, there **is** no instant detection and broadcasting of this emergency. This delay in identifying health problems can lead to preventable casualties because immediate medical assistance is not provided. In many combat scenarios, soldiers are unable to speak or operate radios when injured, which makes manual reporting impossible.

Locating injured soldiers is another complicated issue. Most existing systems use only GPS coordinates to locate soldiers. However, GPS is not always accurate in remote areas, especially in dense forests, inside buildings, or underground. Even if coordinates are available, teammates still struggle to understand how far the injured person is, and in which direction they should move. Every second lost in searching increases the chances of death. There is no mechanism to estimate local distance between soldiers using signal strength, nor is there any directional guidance to reach a soldier quickly. Teams waste precious time trying multiple directions, shouting, or scanning manually for injured comrades, which delays rescue operations.

Another critical battlefield problem is ammunition awareness. Soldiers often do not have a reliable method to know or share how many bullets they have left, especially during stressful combat. Running out of ammunition unexpectedly can cause mission failure, extra risk exposure, or emergency retreat. Commanders have no visibility on individual soldiers' ammunition status, which affects tactical planning and team movement. Existing health monitoring systems completely ignore ammo tracking, even though it is one of the most important real-time battlefield parameters for operational readiness.

Communication overload and noise interference also cause major challenges. In war zones, soldiers face explosions, gunfire, shouting, vehicle engines, and extreme chaos. Audio communication through radios becomes unclear, slow, and unreliable. Soldiers might miss important messages or misinterpret commands. Most wearable monitoring systems do not provide non-audio alert methods such as vibration feedback or visual notifications. If a soldier receives a critical alert, there is no guarantee they will hear it in a high-noise environment, leading to delayed reaction or wrong movement.

Commanders face additional information-based problems. They often do not have a central and updated view of the battlefield situation. They need to know which soldier is healthy, which soldier is injured, which soldier is requesting help, and which soldier still has ammunition available. Without real-time dashboards and connectivity, military decision making becomes slow, uninformed, and reactive rather than proactive. Many existing systems can only display data from one soldier at a time, and data becomes outdated quickly due to communication failures.

Security and confidentiality are also major concerns. Any system using public networks or cloud services is vulnerable to data interception, GPS spoofing, jamming, hacking, and tracking by enemy forces. Leakage of soldier location, health condition, or ammunition status could lead to direct

battlefield disadvantages. Existing approaches do not include secure, closed, peer-to-peer communication mechanisms that work independently without outside servers.

Power management is another practical challenge. Soldiers carry heavy equipment, batteries, and protective gear. Monitoring devices must be low-power, lightweight, and rugged. Many commercially available solutions consume large amounts of energy and require frequent recharging, making them unsuitable in missions that last several hours or days. A monitoring device must continue operating even in harsh conditions like rainfall, dust, high temperature, shock, and vibration, which many existing systems cannot handle. Monitoring of soldier health is another critical problem area. Existing devices may collect vital signs such as heart rate, temperature, or oxygen level, but many of them are not designed for continuous real-time battlefield usage. Some devices only record data and do not send automatic alerts. If a soldier experiences medical issues like low oxygen levels, heat stroke, severe bleeding, heart irregularities, or fatigue, there is no instant detection and broadcasting of this emergency. This delay in identifying health problems can lead to preventable casualties because immediate medical assistance is not provided. In many combat scenarios, soldiers are unable to speak or operate radios when injured, which makes manual reporting impossible.

Overall, the problem lies in the absence of a single, integrated system capable of handling the multiple needs of modern battlefield communications, which include: real-time health monitoring, emergency detection, location tracking, local distance estimation, tactical alerts, and operational data sharing among soldiers without dependency on conventional network infrastructure. This gap leads to compromised mission safety, reduced visibility, uncoordinated response, and ultimately, a higher risk of soldier casualties in critical combat operations.

3.2 Proposed Solution

The proposed solution of our project is to create a wearable and communication system for soldiers that can monitor important information in real time without needing internet, SIM cards, or any external network. Each soldier carries a compact device based on an ESP32 microcontroller. This device has sensors that measure the soldier's heart rate, oxygen level (SpO₂), and body temperature continuously, so that dangerous situations like fainting, dehydration, high fever, stress, or low oxygen can be detected early. A GPS module is added to locate the soldier on a map, which means their movement can be seen live during a mission.

One special feature in our system is ammo tracking, where a button is used to count every bullet shot, telling how many bullets are left in the magazine and how many magazines have been used. This information is very important in a battlefield because soldiers usually have no idea when they are going to run out of bullets, and this can lead to risky situations. The real strength of our solution is the use of ESP-NOW mesh networking. Instead of sending data to a tower or server, each device sends data to the nearest device. If one device is out of range, the data will automatically find another device to reach every soldier in the group. This is called multi-hop communication, and it lets soldiers stay connected even in forests, mountains, bunkers, or enemy areas where network towers are not available.

To make sure soldiers can quickly understand the data, we include a small OLED display on every device. This display shows nearby soldiers, their distance, alerts, ammo count, latest command, and emergency messages. A vibration motor is used to give silent alerts because soldiers might not hear sound in a battlefield filled with noise, explosions, and gunshots. The emergency feature is straightforward. A long button press sends an emergency message to everyone in the group if a soldier is injured or in danger. When another soldier sees the alert, they can accept it with a short button press. The system then shows the direction and distance of the injured soldier using GPS and RSSI signal strength. This means a helper can walk directly toward the injured person without searching blindly or shouting.

All soldier devices constantly share data through the mesh, and one device with Wi-Fi connection sends this live data to a local Node.js WebSocket dashboard. This dashboard acts like a control room screen and shows all soldiers on one display at the same time with their health, location, ammunition level, and emergency status. If someone is injured, it highlights them so that mission control can respond immediately. The dashboard also allows commanders to send messages or instructions back to soldiers, which show up on the OLED screens in the device. This is much faster, simpler, and safer than relying on voice messages in noisy environments.

The proposed solution is designed to be low power, easy to carry, and reliable in extreme environments. Because ESP-NOW consumes less energy than normal Wi-Fi or GSM, the device can run for longer periods on battery. There is no need for cloud servers, SIM cards, or network charges, and this makes the system suitable for difficult terrains like deserts, jungles, borders, and disaster zones. The whole system focuses on solving real battlefield problems such as locating injured soldiers faster, tracking ammunition instantly, monitoring health continuously, and sending silent alerts that are impossible to miss. By combining health sensors, GPS, ammo detection, mesh networking, vibration alerts, and a live dashboard in one unit, the proposed solution gives soldiers and commanders a complete picture of the battlefield. This improves safety, decision-making, teamwork, and mission success, especially in areas where traditional communication systems fail or are not practical.

Another important aspect of our proposed solution is that it works even if one device fails. In most systems, if the main communication device stops working, the whole network stops. But in TACMESH-VITAL, every soldier device can act as a communication bridge for others. This means messages can travel through different paths, and the network can automatically heal itself. If a soldier goes out of range or gets disconnected, data will still travel through nearby soldiers, so the system does not completely fail. Our solution also stores the latest data locally for a short time, so soldiers can still see recent information even if the network is temporarily weak. This increases reliability in real missions where everything is unpredictable. The project is designed to be modular, meaning more soldiers can be added to the mesh without changing the setup. If more troops join the mission, their devices instantly start communicating with existing ones. Because our system uses local communication and not internet servers, it also protects sensitive military data from being leaked or hacked. In addition, the dashboard server runs in a closed local environment, keeping all communication private and secure. Overall, the proposed solution focuses on performance, simplicity, safety, and adaptability, making it highly suitable for real-time field operations.

CHAPTER 4:

METHODOLOGY AND IMPLEMENTATION

4.1 Block Diagram

This diagram illustrates how the TACMESH-VITAL soldier network functions using a combination of local mesh communication and centralized monitoring. The system is built around multiple TacMesh nodes, each based on an ESP32 microcontroller. These nodes communicate with each other directly using ESP-NOW, a peer-to-peer wireless protocol, forming a self-healing mesh network. This means that every device can send, receive, and forward data without needing a router or internet connection. For example, TacMesh Node 1, which contains the vital sensors (MAX30105 for heart rate, SpO₂, and body temperature, along with a GPS module for real-time location), passes health and navigation data to Node 2 and Node 3 using ESP-NOW.

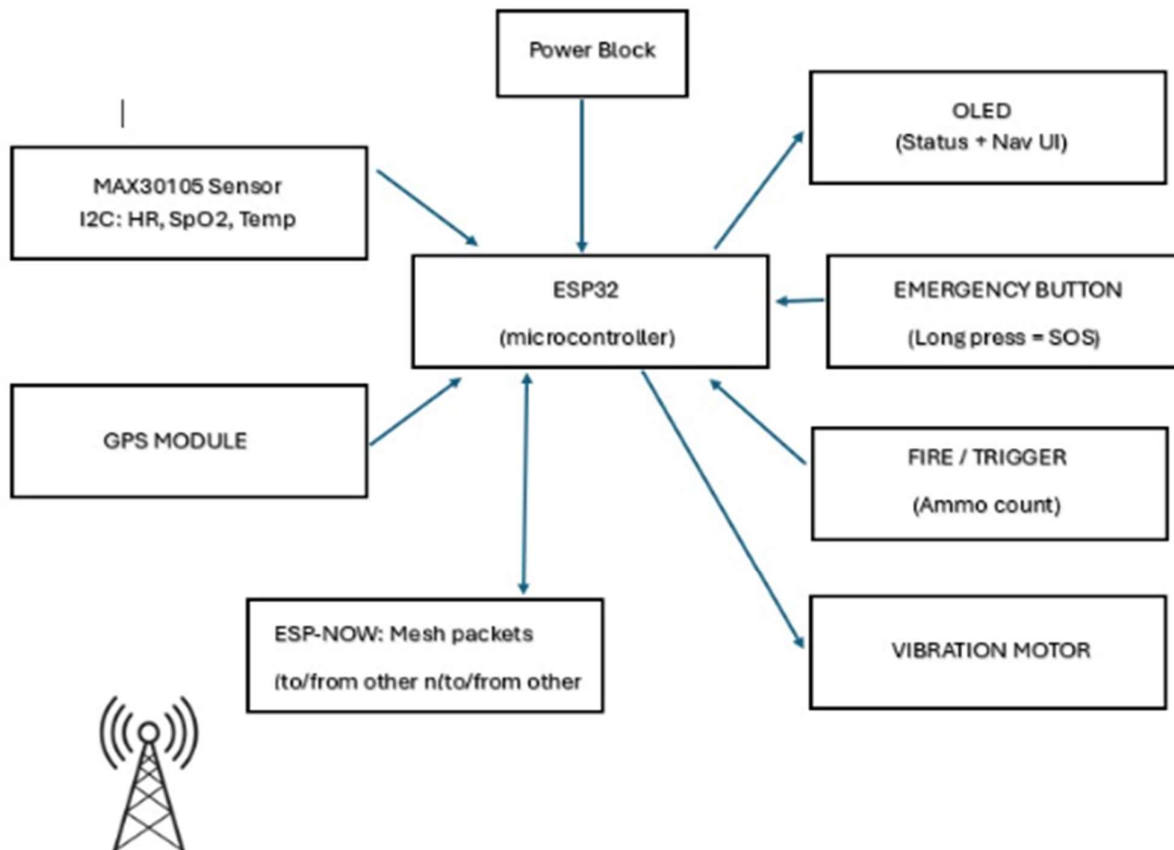


Fig 1. Block Diagram of Tacmesh-Vital: Smart and Secure mesh network for battlefield soldier monitoring and ammo tracking

Similarly, Node 2, which has a vibration motor for tactical alerts, can receive emergency warnings or commands from any other node in range. Node 3 acts as a relay node, extending mesh coverage and ensuring that messages can hop across multiple devices to reach their destination even if one node moves too far or gets disconnected.

In the center of the network, there is a 2.4 GHz Wi-Fi access point (SSID "project2"), which acts as a gateway to the TacMesh Dashboard installed on a desktop computer. Whenever any TacMesh node comes into Wi-Fi coverage, it automatically sends live sensor data to the dashboard using a WebSocket connection, enabling remote real-time visualization. The dashboard displays a live map, vitals chart, ammunition count, node distance relationships, and mesh links. Importantly, the dashboard can also send commands (CMD) back to any soldier. These commands travel through Wi-Fi first and then get relayed through the ESP-NOW mesh, meaning a single connected node is enough to propagate orders across the entire network. This hybrid design makes communication fail-safe, because even if Wi-Fi is unavailable in the battlefield region, nodes can still communicate with each other locally and synchronize data later when one of them reaches a Wi-Fi zone.

Overall, the diagram represents a distributed, secure, and intelligent soldier monitoring system. The mesh topology ensures that no single point of failure exists, while health, location, and tactical status are always updated. Silent vibration alerts reduce noise in combat conditions, and OLED displays allow individual soldiers to view live information without exposing their position. This architecture supports real-time decision-making, team coordination, and safety monitoring in military operations where network infrastructure is unreliable.

This diagram shows how multiple soldier devices communicate with each other and with a central monitoring dashboard using a combination of ESP-NOW mesh communication and normal Wi-Fi (2.4 GHz). Each ESP32 board acts as a TacMesh node, meaning every soldier carries one of these devices.

- **TacMesh Node 1** has sensors such as the MAX30105 (for heart rate, SpO₂, and temperature) and a GPS module to detect location. This node also has an OLED display to show live information to the soldier in the field.

- **TacMesh Node 2** vibration motor has a, which gives silent alerts when there is an emergency warning, command message, or when help is needed.

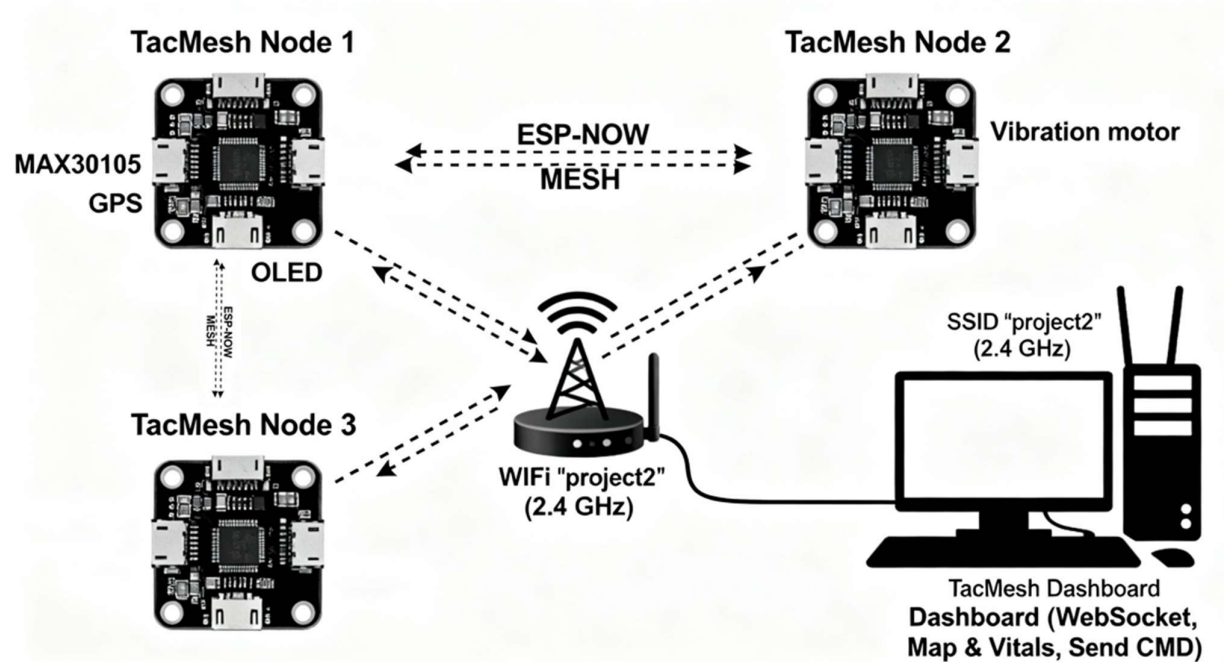


Fig 2 : Block Diagram of Mesh Network

- **TacMesh Node 3** acts as another repeating node in the network, helping to pass messages to other nodes and extend the coverage area.

All nodes communicate with each other using ESP-NOW mesh, represented by dashed arrows. ESP-NOW works without internet and without a router, meaning soldier devices can send and receive data even in remote or battlefield environments where network access is unavailable. Data automatically hops between nodes, so messages can travel across long distances through multiple soldiers acting as relay points.

At the center, there is a normal Wi-Fi access point (SSID "project2") running at 2.4 GHz. If any node has Wi-Fi coverage, it connects to this access point and sends all collected data to the TacMesh Dashboard using a WebSocket connection. This dashboard is installed on a PC and displays:

- live heart rate and SpO₂ readings,
- location data on a map,

- ammunition status,
- emergency alerts,
- and mesh communication status.

The dashboard can also send commands back to any soldier node through ESP-NOW, such as messages like “Move to point”, “Help requested”, “Emergency accepted”, etc.

4.1.1 Pictorial Representation

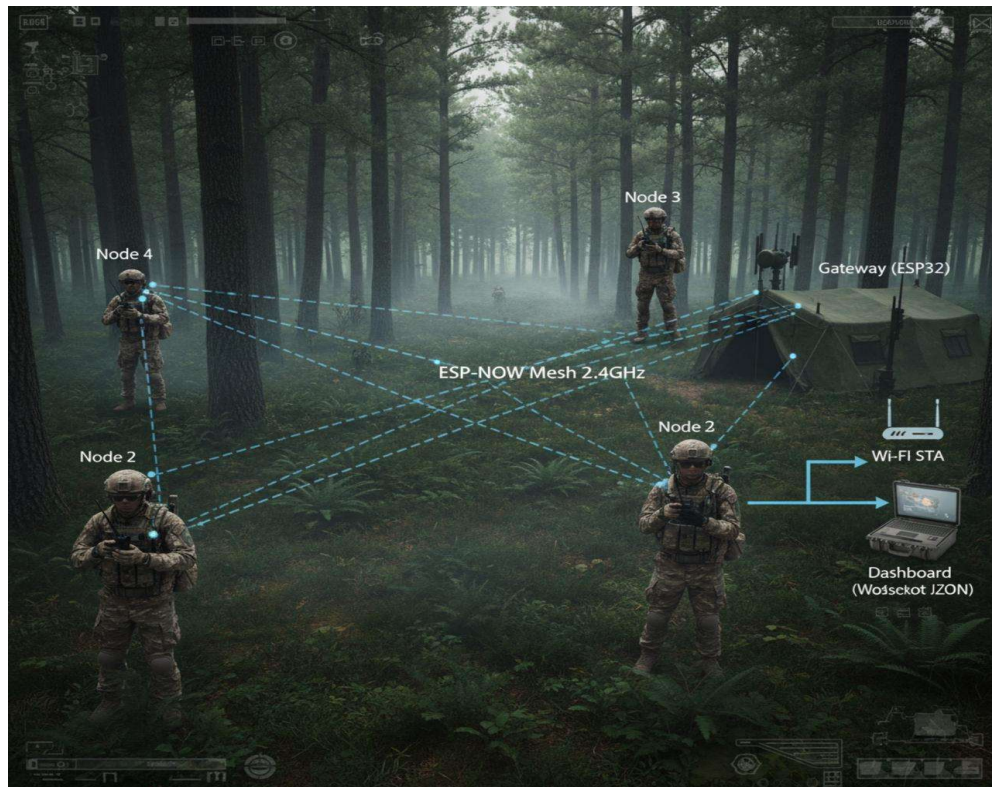


Fig 3: Pictorial Representation of Tacmesh-vital: Smart and secure mesh network for battlefield soldier monitoring and ammo tracking

This image visually demonstrates how the TACMESH-VITAL mesh communication system works in a real battlefield environment. In this scenario, multiple soldiers (Node 1, Node 2, Node 3, Node 4, etc.) are spread across a forest terrain, and each soldier is carrying an ESP32-based TacMesh device. The dashed blue lines between the soldiers represent ESP-NOW 2.4 GHz mesh communication, which allows every node to share data directly with nearby nodes. This creates a wireless network where

information can hop from one soldier to another without needing any mobile network coverage, satellite link, or internet connectivity.

At one side of the field, there is a base camp tent with a special ESP32 device acting as a gateway. This gateway node receives mesh data from the soldiers and connects to a laptop or command center using Wi-Fi STA mode. The laptop runs the TacMesh Dashboard, which uses WebSockets to display real-time information from all nodes. The dashboard shows live vitals (heart rate, SpO₂, temperature), GPS location, emergency status, ammunition tracking, and mesh network activity. This setup allows commanders to view what is happening in the field instantly.

One of the most important aspects of this representation is that every node can communicate with every other node indirectly, even if they are not in direct range. For example, Node 4 can communicate with Node 3 or Node 1 through Node 2 as a relay. This is called a self-healing mesh topology. If one soldier moves behind trees or temporarily loses connectivity, data will still find an alternate route through nearby nodes. This makes communication more reliable in forest, mountain, or urban environments where signals may be blocked.

This pictorial view also clearly shows that the system does not depend on a single communication channel. Soldiers exchange information locally using ESP-NOW mesh, and only one node needs Wi-Fi to forward all the collected data to the dashboard. Even if the Wi-Fi gateway stops working, the soldiers can continue passing messages among themselves using mesh networking. This provides continuous situational awareness, enhanced safety, and low-power communication suitable for tactical missions.

4.1.2 Structure

1. ESP32 (Main Controller)

- The ESP32 is the main microcontroller that reads sensor values, handles button inputs, and controls the display and vibration motor.
- It also builds and sends structured data packets containing vitals, GPS, ammo, and emergency status.
- Using ESP-NOW wireless protocol, the ESP32 communicates directly with other nodes without internet, SIM, or router

2. MAX30105 Sensor (Vitals Monitoring)

- The MAX30105 continuously measures heart rate, SpO₂ (oxygen level), and temperature using infrared and red LED light absorption.
- These values help detect stress, fatigue, low oxygen levels, or possible injury in the soldier.
- The ESP32 processes sensor readings periodically and updates the display and mesh packets.

3. GPS Module (Location Tracking)

- The GPS module continuously receives signals from satellites to calculate latitude and longitude.
- This helps identify the exact position of each soldier in real time.
- The location data is useful for navigation, locating injured soldiers, and command center mapping.

4. Emergency Button

- The emergency button is used to trigger an SOS distress signal during critical situations.
- When long pressed, it automatically broadcasts an alert to all nearby nodes via ESP-NOW mesh.
- Other soldiers can see which node is in trouble and respond immediately.

5. Trigger Button (Ammo Counter)

- The trigger button records a shot every time it is pressed, simulating live ammunition use.
- The system tracks bullets remaining in the magazine and total number of shots fired.
- This prevents unexpected ammo shortage and supports better tactical planning.

6. OLED Display

- The OLED screen shows all essential data to the soldier in a compact format.
- It displays heart rate, oxygen level, temperature, ammo, GPS distance, emergency messages, and commands.
- This enables real-time decision-making and awareness without external communication devices.

7. Vibration Motor (Silent Alerts)

- The vibration motor provides haptic (touch-based) feedback using different patterns.
- Emergency alerts, low ammo, or received commands are indicated silently.
- This is extremely useful during stealth missions where sound-based alerts are risky.

8. ESP-NOW Mesh Communication

- ESP-NOW creates a distributed wireless network where each ESP32 node communicates directly with nearby nodes.
- Data hops from node to node, extending communication range without a base station.
- This mesh communication is fast, encrypted, low-power, and reliable in outdoor environments

9. Wi-Fi Gateway Node

- Only one node needs to access Wi-Fi to send and receive data from the command dashboard.
- It collects all mesh data and forwards it using WebSockets to the monitoring PC.
- This hybrid design allows offline mesh networking + online remote monitoring at the same time.

10. Dashboard / Control Center

- The dashboard interface shows multiple soldiers in one window with live updates.
- Commanders can observe vitals, GPS location, ammo count, and emergency status simultaneously.
- They can also send commands or instructions back to any soldier, improving coordination and safety.

4.1.3 Flow chart

The flowchart illustrates the complete working process of the TACMESH-VITAL system, starting from power-on to collecting sensor data, forming mesh communication, and sending information to the dashboard. It shows how the ESP32 node behaves under different network conditions and how real-time data moves through the system. The flow begins with the START point, which tells us that the program has just been powered on or reset.

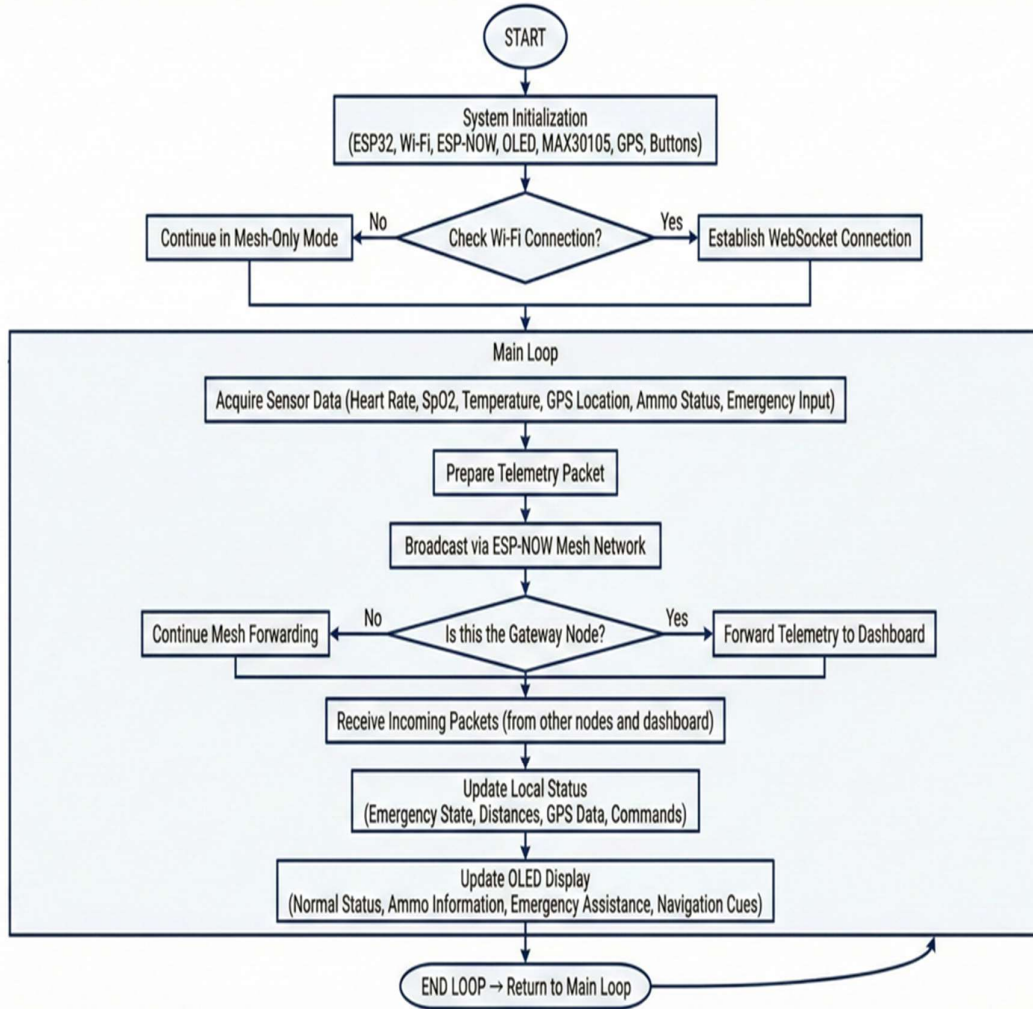


Fig 4: Flow chart of secure mesh network for soldier monitoring and ammo tracking

1. System Initialization

The first major process is the System Initialization, where the ESP32 sets up all required hardware and communication modules. At this stage, the system activates and configures Wi-Fi, ESP-NOW mesh protocol, OLED display, MAX30105 sensor, GPS module, and all button inputs. This ensures that every component is ready before real-time operations begin.

2. Check for Wi-Fi Connection

After initialization, the system checks if Wi-Fi is available. If Wi-Fi is not available, the ESP32 automatically switches to mesh-only operation, meaning it will not attempt to reach the dashboard directly but will still communicate with other nodes using ESP-NOW. If Wi-Fi is available, the system

establishes a WebSocket connection with the dashboard. This hybrid approach allows the device to operate both online and offline depending on field conditions, which is very useful in military or remote applications.

3. Main Loop Operation

After completing the initial configuration, the system enters its Main Loop, which continuously runs until the device is powered off. This loop includes several important sequential tasks.

4. Sensor Data Acquisition

The first step in the main loop is to acquire real-time sensor data. The ESP32 reads the soldier's vital parameters such as heart rate, oxygen level (SpO₂), body temperature, GPS coordinates, ammunition count, and emergency button status. This step is important because all further decisions and communication depend on accurate, up-to-date sensor readings.

5. Prepare Telemetry Packet

After gathering data, the system combines all information into a telemetry packet. A packet is a structured data package that includes readings, node ID, time stamp, emergency status, GPS data, and mesh routing details. Preparing a packet makes communication consistent and organized, so that any node receiving it can easily understand the information.

6. Broadcast via ESP-NOW Mesh

The next step is to broadcast the telemetry packet through ESP-NOW. Broadcasting means sending the packet so it can be received by nearby nodes. ESP-NOW does not require the internet or an access point, allowing multiple ESP32 nodes to form a mesh network. This ensures every soldier device can communicate peer-to-peer even in remote areas.

7. Gateway Check

After broadcasting, the system checks a very important condition: Is this node acting as the gateway? A gateway node is responsible for forwarding mesh data to the Wi-Fi dashboard.

- If **Yes**, the telemetry is forwarded directly to the dashboard through WebSocket.

- If **No**, the node continues normal mesh forwarding, meaning it helps relay the packet to other nodes in the network.

This logic makes the network scalable. Only one node needs Wi-Fi access, while the others function wirelessly in mesh mode.

8. Receive Incoming Packets

Every node also listens for incoming packets from other nodes and the dashboard. This makes the system bidirectional, meaning it can send and receive data. Incoming packets can contain:

- Other soldier vital data
- Emergency alerts
- Commands from dashboard
- GPS information
- Navigation instructions

This step creates real-time interaction across the system.

9. Update Local Status

Once a packet is received, the node updates its local status memory. This includes storing:

- Emergency status of any soldier
- Distance estimates between nodes
- GPS coordinates
- Ammo information
- Received command messages

Keeping local status updated allows quick decision-making even without dashboard access.

10. Update OLED Display

After processing data, the system updates the OLED display so the soldier can see current information. Different modes may be displayed:

- Normal mode: vitals, ammo, GPS

- Emergency mode: rescue arrow, target distance
- Command mode: received instructions

The vibration motor may also activate at this stage if alerts or new commands are received.

11. End Loop and Repeat

Finally, the flowchart reaches END LOOP → Return to Main Loop, meaning the cycle repeats continuously. New data is collected every loop, and communication happens continuously in real-time. The loop can run hundreds of times per second depending on sensor availability.

In this way, the flowchart clearly represents how TACMESH-VITAL maintains continuous communication and monitoring in real time. Each stage of the loop ensures that vital parameters are updated, packets are shared among nodes, and the soldier interface is refreshed with the latest information. This continuous cycle makes the system fast, reliable, and adaptable in different field conditions without depending on any central server. This flowchart also highlights the adaptive nature of the system. Even if Wi-Fi becomes unavailable, communication does not stop because the devices automatically switch to mesh-only mode. This self-healing design ensures that no single node failure breaks the entire network. By repeatedly collecting data, forwarding packets, and updating displays, TACMESH-VITAL keeps soldiers informed and commanders aware of every critical situation. Overall, the flowchart shows a complete real-time cycle of sensing, decision-making, transmission, and feedback that makes the project robust, efficient, and suitable for actual defense applications.

CHAPTER 5:

RESULT AND DISCUSSION

We implemented and tested using multiple ESP32 nodes equipped with MAX30105 sensors, GPS modules, and vibration alerts. During testing, the system was able to continuously collect and transmit vital parameters such as heart rate, SpO₂, and body temperature from each soldier node. The data transmission worked reliably through ESP-NOW mesh networking even in areas without Wi-Fi coverage. Every node received live updates from nearby nodes, and multi-hop communication proved effective as data traveled from one node to another without any noticeable delay.



Fig 4: Prototype wearable glove setup with sensors and ESP32 connections.

The emergency alert feature worked efficiently. By using a long press of the emergency button, a node could broadcast an SOS message that appeared on all other nodes' OLED displays. The vibration motor provided silent, instant alerts, which ensured that signals could be recognized even in noisy environments. A second node could accept the emergency by pressing its button, which changed the

alert status and helped identify the designated helper. GPS readings were successfully used to calculate distance and direction toward the emergency node, making it easier to locate injured personnel. The system's RSSI-based distance estimation also showed practical results for close-range navigation.

The ammo tracking system showed clear results. Each press of the trigger button successfully counted shots fired and displayed bullets remaining in the current magazine. When the magazine count reached zero, the device alerted the user that a reload was needed through vibration feedback. This function proved reliable and easy to understand during field tests.

The real-time dashboard allowed clear visualization of collected data through WebSockets. Live values for vital signs, ammunition count, location, and emergency status appeared instantly on the command center screen. The dashboard also displayed node-to-node distance connections, which helped analyze network topology and communication performance. Command messages sent from the dashboard were received by the selected ESP32 nodes, and the vibration feedback confirmed message reception.

Overall, test results showed that ESP-NOW mesh networking is a suitable, low-power, and fast communication method for soldier-to-soldier data transfer. The hybrid setup (mesh + Wi-Fi gateway) provided flexibility by enabling offline operation in the field and online monitoring in a control room. The system worked without the need for cellular networks, internet connectivity, or cloud servers, making it highly reliable in remote and tactical environments. These results indicate that TACMESH-VITAL has strong potential to improve battlefield safety, coordination, and response times by combining health monitoring, ammunition awareness, and emergency alerts into a single integrated wearable system.

CHAPTER 6:

FUTURE TRENDS AND CONCLUSION

6.1 CONCLUSION

The TACMESH-VITAL project presents a practical and innovative solution for real-time soldier monitoring and battlefield communication where normal networks are weak or unavailable. By integrating vital sign sensing, GPS tracking, emergency messaging, and ammunition counting into a wearable unit, the system proves that multiple critical data points can be collected and shared wirelessly using minimal hardware.

A major strength of TACMESH-VITAL is the use of ESP-NOW mesh networking, which enables node-to-node communication without mobile data, satellite, or cloud dependence. This ensures reliable data sharing in forests, mountains, deserts, and war zones where standard communication systems fail. The multi-hop communication method allows data to reach all team members even if some nodes are far away, while hybrid connectivity ensures that only one device needs Wi-Fi to update a central command dashboard when available.

Health monitoring using the MAX30105 sensor provides real-time readings of heart rate, SpO₂, and temperature, helping detect fatigue, stress, blood loss, and breathing issues. GPS data improves situational awareness and supports quick rescue during emergencies. The emergency button and vibration system allow silent alerts suited for tactical and stealth missions. Ammunition monitoring improves planning and reduces the risk of running out of bullets in critical situations.

Prototype testing showed low-latency transfers, stable mesh data relay, fast emergency alerts, and smooth dashboard visualization. Using low-cost microcontrollers and open-source tools makes the solution affordable, scalable, and suitable for practical deployment.

In conclusion, TACMESH-VITAL demonstrates that a decentralized, secure, and smart battlefield support system can be built using IoT technology. It improves soldier safety, enhances coordination, and supports faster decisions during operations. With future upgrades such as encryption, PCB miniaturization, solar power, LoRaWAN range extension, and AI-based health prediction, this system can become a strong next-generation defense monitoring tool.

6.2 Future Trends

1. Miniaturized PCB Design

All sensors, buttons, and the ESP32 can be integrated into a single compact PCB to reduce wiring, weight, and power consumption.

2. Solar-Powered Wearable Nodes

Adding flexible solar panels on the uniform or backpack to continuously charge the system during missions.

3. Encrypted Communication

Implementing end-to-end AES or military-grade encryption for secure data transfer over ESP-NOW and Wi-Fi.

4. LoRa + Mesh Hybrid Network

Integrating LoRa long-range technology to extend communication beyond 5 km while still keeping mesh features.

5. AI-Based Health Prediction

Using machine learning to detect stress, dehydration, shock, or abnormal heart activity automatically.

6. Cloud Data Storage and History Logs

Saving soldier data in a secure cloud server for long-term monitoring, training improvement.

7. Wearable Body Integration

Converting glove prototype into a wearable vest, armband, or smart jacket for better durability and comfort.

8. Automatic Location Guidance

Integrating map routing and arrow navigation on OLED display to guide soldiers to injured teammates automatically.

9. Integration with UAV/Drone Support

Sending emergency locations directly to drones for fast aerial assistance, surveillance, or supply drop missions.

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